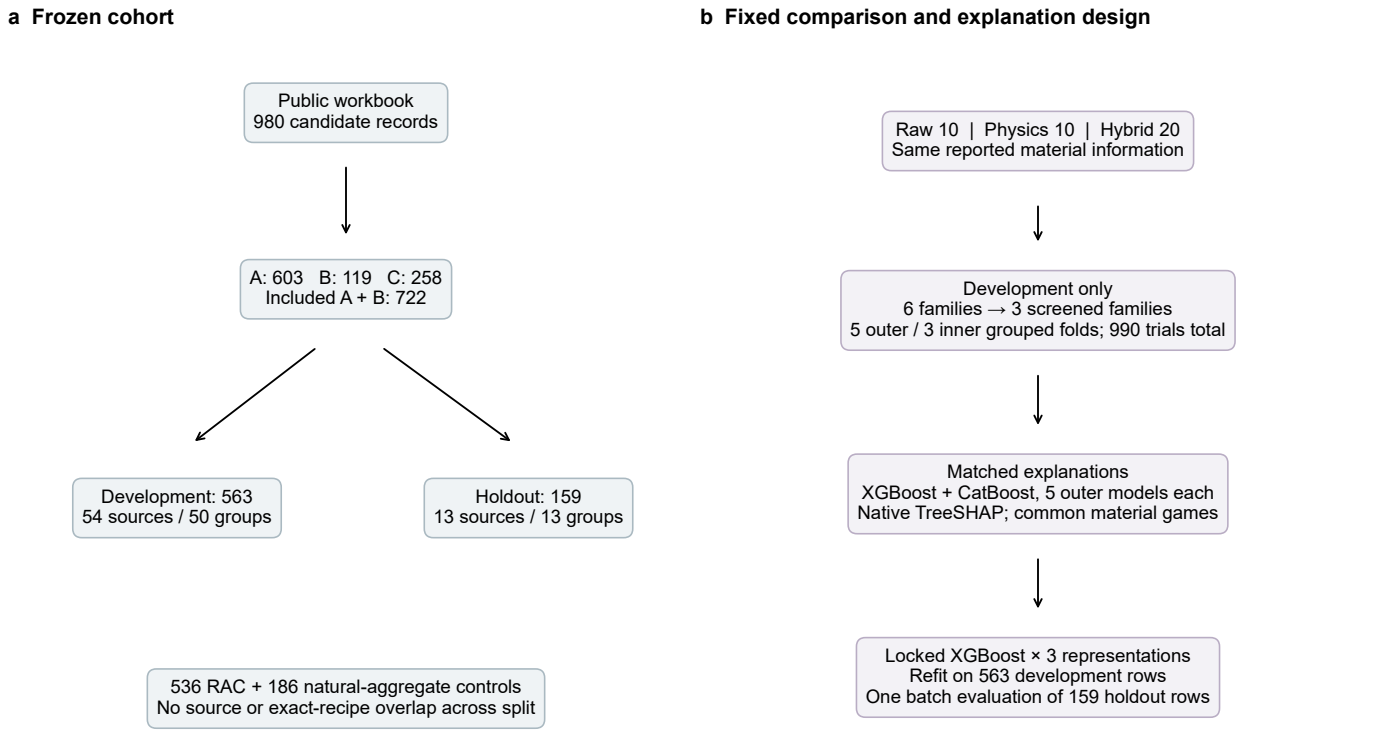

a Frozen cohort
Public workbook
980 candidate records
A: 603 B: 119 C: 258
Included A + B: 722
Development: 563
54 sources / 50 groups
Holdout: 159
13 sources / 13 groups
536 RAC + 186 natural-aggregate controls
No source or exact-recipe overlap across split
b Fixed comparison and explanation design
Raw 10 | Physics 10 | Hybrid 20
Same reported material information
Development only
6 families → 3 screened families
5 outer / 3 inner grouped folds; 990 trials total
Matched explanations
XGBoost + CatBoost, 5 outer models each
Native TreeSHAP; common material games
Locked XGBoost × 3 representations
Refit on 563 development rows
One batch evaluation of 159 holdout rows